PRODUCT REQUIREMENTS DOCUMENT (PRD)

LOCAL-FIRST FINANCIAL ANALYSIS APPLICATION (V1)

---

1. PRODUCT OVERVIEW

1.1 Product Summary

The product is a local-first macOS desktop application designed for financially sophisticated individual investors who want deep access to structured company financial data derived primarily from SEC filings.

The application functions conceptually as a lightweight fundamentals-oriented Bloomberg terminal without any cloud-hosted backend infrastructure. All company data is ingested on demand, processed locally, and persisted entirely on the user's machine.

The primary value proposition is:

- High-confidence financial data accuracy
- Deep historical coverage
- Full offline usability after ingestion
- Full traceability back to SEC filings
- Zero SaaS dependency

The system prioritizes correctness, auditability, and transparency over ingestion speed or flashy UI interactions.

1.2 Product Vision

Enable sophisticated investors to independently analyze public companies using authoritative SEC-derived structured data without relying on cloud services, subscription terminals, or opaque third-party financial aggregators.

1.3 V1 Product Scope

V1 focuses on:

- Single-company analysis
- Historical financial statements
- Derived metrics using fixed formulas
- Time-series visualization
- Local persistence
- Manual ingestion and refresh
- Offline-first workflows
- SEC traceability

V1 intentionally excludes collaborative, social, predictive, AI-assisted, or cloud-hosted capabilities.

1.4 Platform Scope

Primary target platform:

- macOS desktop

Acceptable implementation approaches:

- Native macOS application
- Electron/Tauri desktop wrapper
- Hybrid desktop architecture

The application must behave as a true desktop application with reliable local persistence and offline functionality.

---

2. GOALS AND NON-GOALS

2.1 Goals

Core Goals

1. Deliver highly accurate financial statement data derived from SEC filings

2. Provide full offline usability for previously ingested companies

3. Enable historical analysis across long time periods
4. Support sophisticated financial workflows without educational simplification
5. Provide traceability for every displayed financial datapoint
6. Establish a modular architecture for future analytical expansion

---

2.2 Non-Goals (V1)

The following are explicitly excluded from V1:

Analytical Features

- Multi-company comparison
- Peer benchmarking
- Screening
- Quantitative ranking systems
- Portfolio management
- Valuation modeling
- DCF generation
- AI-generated insights
- Earnings call analysis

Data Features

- Real-time quotes
- Intraday market data
- News feeds
- Insider trading feeds
- Analyst estimates
- Alternative data

Collaboration Features

- User accounts
- Cloud sync
- Shared dashboards
- Multi-device synchronization

Customization Features

- User-defined formulas
- Plugin ecosystem
- Custom widgets *(V1 non-goal; delivered in V2 as user-imported custom metrics — see `docs/custom-metrics-prd.md`)*

Export Features

- CSV export *(V1 non-goal; delivered in V2 — see `docs/custom-metrics-prd.md`)*
- PDF export
- Image export
- Spreadsheet integrations

---

3. TARGET USERS

3.1 Primary Users

Sophisticated Individual Investors

Characteristics:

- Comfortable reading financial statements
- Familiar with accounting terminology
- Understand valuation concepts
- Prefer raw structured data over simplified summaries
- Require transparency into calculations

3.2 Secondary Users

Financial Researchers

Individuals performing long-horizon company analysis who value:

- Historical consistency
- Offline accessibility
- Reproducibility
- Auditability

3.3 User Expectations

Users expect:

- High data accuracy
- Transparent calculations
- Stable historical records
- Minimal abstraction
- Fast navigation after ingestion
- Trustworthy financial lineage

Users do NOT expect:

- Beginner education
- Gamification
- Social interaction
- AI-generated narratives
- Brokerage integration

---

4. USER PERSONAS

4.1 Persona A — Long-Term Fundamental Investor

Background

- Tracks 20–50 companies
- Performs deep annual analysis
- Values long historical trends
- Uses SEC filings directly today

Pain Points

- Existing tools are expensive
- Data providers disagree
- SEC filings are cumbersome to navigate manually
- Offline workflows are difficult

Goals

- Quickly inspect long-term company fundamentals

- Validate financial numbers directly against filings
- Analyze financial trends without subscriptions

4.2 Persona B — Independent Financial Researcher

Background

- Builds personal investment theses
- Wants reproducible datasets
- Distrusts opaque aggregators

Pain Points

- Hard to verify third-party financial data
- Derived metrics often lack traceability

Goals

- Inspect source lineage for all metrics
- Build confidence in extracted data
- Retain complete local control over datasets

---

5. USER WORKFLOWS

5.1 Initial Company Ingestion Workflow

Flow

1. User launches application
2. Home screen loads saved companies
3. User enters ticker symbol
4. System validates ticker
5. System downloads SEC filing metadata
6. System identifies relevant filings
7. System downloads filings
8. System parses structured filing data
9. System normalizes financial data
10. System stores normalized data locally

11. Dashboard becomes available

5.2 Company Analysis Workflow

Flow

1. User selects saved company
2. Dashboard loads local data
3. User views:
    - Summary widgets
    - Financial tables
    - Charts
    - Derived metrics
4. User drills into metrics
5. User views filing traceability
6. User switches between annual and quarterly views

5.3 Refresh Workflow

Flow

1. User presses refresh button
2. System checks for newer filings
3. System downloads updated filings
4. System re-processes affected data
5. Local database updates
6. Dashboard refreshes

5.4 Offline Workflow

Flow

1. User launches app without internet
2. Previously ingested companies remain accessible
3. All dashboards, charts, and tables function normally
4. Refresh and new ingestion operations disabled

---

6. FUNCTIONAL REQUIREMENTS

6.1 Company Management

Requirements

FR-001 — Add Company

Users must be able to add companies by ticker symbol.

FR-002 — Persist Saved Companies

Saved companies must persist across application restarts.

FR-003 — Remove Company

Users must be able to remove companies from saved list.

FR-004 — Cached Historical Data

Removing a company may optionally preserve cached data locally.

6.2 Financial Data Retrieval

Requirements

FR-010 — SEC Filing Discovery

System must identify:

- 10-K filings
- 10-Q filings
- 10-K/A and 10-Q/A amendments
- 8-K filings under Item 4.02 (Non-Reliance on Previously Issued Financial Statements)

8-K filings under Item 4.02 are not used as a source of financial data. They are ingested solely so that the system can:

1. Identify the specific fiscal periods the disclosure flags as unreliable, with full accuracy regardless of phrasing or whether structured tags are present.
2. Surface a per-period warning to the user that the displayed values for those periods may be unreliable.
3. Clear the warning automatically once a corresponding 10-K/A or 10-Q/A covering the affected period(s) has been filed and ingested.

All other 8-K item types are out of V1 scope.

FR-011 — Structured Filing Preference

System must prioritize:

1. XBRL/XML structured data
2. Structured SEC APIs
3. HTML/text parsing fallback

FR-012 — Historical Depth

System must retrieve:

- Minimum 10 years
- Target 20 years

FR-013 — Quarterly Coverage

Quarterly data must be supported where available.

6.3 Financial Statements

Requirements

FR-020 — Income Statement Support

Must support:

- Revenue
- Gross profit
- Operating income
- Net income
- EPS
- Shares outstanding

FR-021 — Balance Sheet Support

Must support:

- Cash
- Debt
- Assets
- Liabilities
- Equity

FR-022 — Cash Flow Statement Support

Must support:

- Operating cash flow
- Capital expenditures
- Depreciation/amortization
- Free cash flow inputs

6.4 Derived Metrics

Requirements

FR-030 — Fixed Derived Metrics

V1 supports predefined formulas only. Required derivations include:

- **Free cash flow** (see FR-032).
- **Operating margin** (see FR-033).
- **Total debt** = `long_term_debt + current_debt`, computed at read time so each input's most-recent primary value is used.
- **Gross profit** = `revenue − cost_of_revenue`, used when the issuer does not directly tag a gross-profit value.
- **Capital expenditures fallback**: when no explicit cash-flow CapEx value is reported (some bank/financial filers omit it), derive from the property, plant and equipment roll-forward — `capital_expenditures = ΔPP&E_net + depreciation_and_amortization` — using the same period's PP&E delta and depreciation. Lineage records both inputs.

- **Bank revenue fallback**: when the issuer does not directly tag `us-gaap:Revenues` (typical for bank-holding companies), derive in the following order until a complete result is available:

    1. `NetInterestIncome + NoninterestIncome`
    2. `(InterestIncomeOperating − InterestExpense) + NoninterestIncome`

    Each derivation writes lineage naming the contributing concepts.

FR-031 — Formula Transparency

Users must be able to inspect formulas used.

FR-032 — Free Cash Flow

Must support free cash flow computation using a predefined formula, derived at read time per period (so each input uses its most-recent primary value):

```
free_cash_flow = net_income + depreciation_and_amortization − capital_expenditures
```

Sign convention: capital expenditures is stored as a positive cash outflow (sign-normalized per §6.2), so it is subtracted; depreciation and amortization is a positive non-cash add-back. All three inputs are required for a period — a period missing any input is omitted rather than reported with an input silently treated as zero. Capital expenditures uses the same read-time series as FR-030 (including the property-plant-and-equipment roll-forward fallback), so free cash flow is available for filers that do not tag cash-flow capital expenditures directly.

FR-033 — Operating Margin

Must support operating margin as a predefined derived metric, computed at read time per period:

```
operating_margin = operating_income ÷ revenue
```

Stored as a dimensionless decimal ratio in micro-units (ratio × 1,000,000, per the §6.2 convention for pure ratios); for example a 25.3% margin is `253000`. Revenue uses the read-time revenue series (including the bank-revenue fallback). Both inputs are required; a period whose revenue is non-positive is omitted (the margin is undefined). An operating loss yields a valid negative margin and is preserved. The UI renders the value as a percentage.

FR-034 — Free Cash Flow Yield

Must support free cash flow yield as a predefined derived metric, computed at read time per period:

```
free_cash_flow_yield = free_cash_flow ÷ market_cap
```

Stored as a dimensionless decimal ratio in micro-units (ratio × 1,000,000, per §6.2); a 4.0% yield is `40000`. The UI renders it as a percentage.

- **Annual** periods use the annual free cash flow.
- **Quarterly** periods use **trailing-twelve-month** free cash flow — the sum of the quarter and its three predecessors — so the series is an annualized, comparable yield rather than a seasonal single-quarter figure. A quarter without four strictly-consecutive trailing quarters is omitted (no partial trailing-twelve-month value).

The denominator is the period's market cap (FR-050). Both numerator and market cap are required; a non-positive market cap is omitted (yield undefined). A cash-burning period yields a valid negative number, which is preserved.

Trailing-twelve-month free cash flow is itself exposed as a derived metric (`free_cash_flow_ttm`) so the quarterly yield is transparent.

6.5 Financial Data Normalization

Requirements

FR-040 — Financial Data Normalization Layer

The system must normalize raw SEC filing data into a consistent internal financial representation suitable for accurate analysis, visualization, historical comparison, storage, and derived metric computation.

Normalization is a core architectural requirement of the platform and is essential to ensuring consistency across companies, filing periods, filing formats, and reporting conventions.

The normalization layer must address inconsistencies and variations including, but not limited to:

- Differences in XBRL taxonomy usage across companies
- Different labels for equivalent financial concepts
- Changes in company reporting conventions over time
- Unit inconsistencies (dollars, thousands, millions)
- Fiscal calendar differences
- Quarterly versus year-to-date reporting formats
- Sign convention inconsistencies
- Amended or restated filings

- Missing or partially structured data
- Filing-format differences between XML/XBRL and text-based filings

Examples:

- Revenue
- SalesRevenueNet
- NetSales

may all map internally to a canonical metric representation such as:

- revenue

The normalization layer must preserve traceability to the original filing source while simultaneously exposing a stable internal schema to downstream systems including:

- Dashboard widgets
- Charts
- Financial tables
- Derived metric engines
- Historical trend analysis
- Future comparison engines

The application must never silently discard normalization conflicts or ambiguities. Any unresolved inconsistencies should be surfaced transparently to the user or recorded in ingestion diagnostics.

Accuracy and consistency are prioritized over ingestion speed.

6.6 Dashboard and Visualization

Requirements

FR-050 — Summary Widgets

Dashboard must display:

- Revenue
- Net income
- Free cash flow
- Free cash flow yield (FR-034)
- Historical market cap per period (see below)
- Total debt

- Derived metrics

Historical market cap is computed at **read time** per period as `close_price(period end date) × shares_outstanding_basic(period)`. End-of-day closing prices are fetched once at ingestion from the market-data adapter (§7.5) — one per distinct period end-date, resolved to the nearest prior trading day — and persisted in `historical_price`, so the market-cap series is available offline once a company has been ingested. (Market cap is derived at read time, like total debt and free cash flow, rather than persisted as a value, so it never goes stale when shares outstanding is superseded; only the external price is persisted.) A period whose price could not be fetched simply has no market cap, and hence no free-cash-flow yield, until a later refresh succeeds — this never affects any other dashboard functionality.

> **Revision note.** An earlier draft specified market cap at each *filing* date, persisted as a value. It was changed to **period-end** price (filing lags vary across companies, so period-end is more comparable) computed at **read time** (supersession-safe), consistent with the other read-time derivations. Current (live) market cap remains deferred.

FR-051 — Historical Charts

Charts must support:

- Annual mode
- Quarterly mode

FR-052 — Financial Tables

Tables must display:

- Historical periods
- Raw line items
- Drill-down details

FR-053 — Drill-Down Navigation

Users must be able to navigate into supporting line items.

6.7 Traceability

Requirements

FR-060 — Source Filing Traceability

Every datapoint must map to:

- Filing
- Filing type
- Filing date

FR-061 — Source Location Traceability

System should retain:

- XBRL concept
- Filing section
- Line references where feasible

FR-062 — Source Preference Visibility

UI must indicate whether value originated from:

- XBRL/XML
- Parsed text
- External API

---

7. NON-FUNCTIONAL REQUIREMENTS

7.1 Reliability

- Local data persistence must be durable
- Corruption-resistant storage required
- Application crashes must not destroy datasets

7.2 Offline Capability

- Previously ingested companies fully accessible offline
- No cloud dependency for analysis workflows

7.3 Maintainability

Architecture must support:

- Additional financial metrics
- Additional data sources
- Future comparison features

7.4 Extensibility

System should support future:

- Formula engines
- Peer analysis
- Export systems
- Plugin systems

7.5 Observability

System should provide:

- Ingestion logs
- Parsing diagnostics
- Data lineage metadata

---

8. DATA MODEL OVERVIEW

8.1 Core Entities

Company

Fields:

- Ticker
- CIK
- Company name
- Exchange

Filing

Fields:

- Filing ID
- Filing type
- Filing date
- Source URL
- Raw filing path

Financial Period

Fields:

- Fiscal year
- Fiscal quarter
- Start date
- End date

Financial Metric

Fields:

- Metric name
- Value
- Currency
- Period
- Filing source
- Source confidence

Derived Metric

Fields:

- Formula ID
- Inputs
- Computed value
- Computation timestamp

8.2 Storage Recommendations

Recommended local storage:

- SQLite
- DuckDB
- Embedded relational database

Raw filing storage:

- Local filesystem

---

9. SYSTEM CONSTRAINTS

9.1 No Backend Infrastructure

System cannot require:

- Cloud storage
- Central APIs
- Hosted databases

9.2 SEC Dependency Constraints

SEC filings may:

- Change formatting
- Contain inconsistent XBRL tags
- Include amended filings

System must tolerate these inconsistencies.

9.3 API Constraints

Yahoo/Google finance APIs may:

- Rate limit
- Change response formats
- Become unavailable

System must gracefully degrade.

---

10. UI/UX REQUIREMENTS

10.1 Design Philosophy

The UI should feel:

- Dense but readable
- Professional
- Analytical
- Efficient
- Minimalistic

Avoid:

- Excessive animations
- Consumer-style simplifications
- Educational overlays

10.2 Home Screen

Must include:

- Saved companies list
- Search/add ticker
- Recent refresh status

10.3 Dashboard Layout

Recommended layout:

- Top summary widgets
- Middle chart section
- Lower financial tables

10.4 Loading States

Must support:

- Ingestion progress
- Parsing progress
- Refresh progress

10.5 Error Visibility

Users must clearly see:

- Missing data
- Parsing failures
- Partial ingestion issues

---

11. DATA INGESTION AND PROCESSING

11.1 Ingestion Pipeline

Step 1 — Filing Discovery

Discover relevant SEC filings.

Step 2 — Filing Download

Download:

- XBRL
- XML
- Filing metadata

Step 3 — Parsing

Extract:

- Financial concepts
- Time periods
- Units
- Filing context

Step 4 — Normalization

Normalize:

- Fiscal periods
- Units
- Metric naming
- Sign conventions
- Filing taxonomy differences
- Reporting inconsistencies

The normalization stage is considered a critical subsystem of the platform rather than a lightweight transformation step.

Step 5 — Persistence

Store:

- Raw filings
- Normalized metrics
- Traceability metadata

11.2 Data Normalization Challenges

System must handle:

- Different fiscal calendars
- Renamed XBRL concepts
- Stock splits
- Restatements
- Currency inconsistencies
- Partial filings
- Conflicting filing representations
- Quarterly versus cumulative reporting differences (single-quarter and year-to-date facts may share the same `fp` tag and are distinguishable only by the duration of the reported period)
- SEC `companyfacts` `fy` / `fp` tags reflect the FILING's fiscal year, not the period the fact represents. A 10-K's embedded comparative data (prior two fiscal years) is tagged with the latest filing's year. The system must derive each period's fiscal year from the period-end date against the issuer's fiscal-year-end calendar rather than trusting the SEC tag, otherwise three years of data collapse into a single slot.
- Within a single (canonical metric, fiscal year) series, a company may file the metric under multiple XBRL concepts whose scopes differ (e.g. `DepreciationAndAmortization` annual-only vs `DepreciationAmortizationAndAccretionNet` quarterly-with-accretion). Mixing concepts when deriving a single quarter from cumulative inputs (e.g. Q4 = FY − 9M) yields a wrong value. The system must select one source concept per (metric, fiscal year) and use only that concept for derivations.
- Banks report revenue as net interest income plus non-interest income rather than via `us-gaap:Revenues`. The system must support a documented fallback chain so bank revenue is not silently zero.

11.3 Filing Priority Logic

Priority order:

1. Latest amended filing
2. Latest original filing
3. Structured data
4. Text fallback

---

12. ERROR HANDLING AND EDGE CASES

12.1 Missing Data

If data unavailable:

- Show explicit gaps
- Avoid fabricated estimates

12.2 Inconsistent Filings

If filings conflict:

- Prefer latest amendment
- Preserve audit trail

12.3 Partial Parsing Failures

System should:

- Continue ingesting valid sections
- Mark incomplete areas

12.4 Offline Errors

When offline:

- Disable refresh
- Preserve dashboard functionality

12.5 API Failures

If supplemental APIs fail:

- Continue SEC-based functionality
- Mark market data unavailable

13. PERFORMANCE CONSIDERATIONS

13.1 Performance Priorities

Priority order:

1. Accuracy
2. Stability
3. Traceability
4. Speed

13.2 Expected Latency

Acceptable:

- Multi-minute initial ingestion
- Noticeable refresh delays

Not acceptable:

- UI freezing
- Data corruption
- Silent failures

13.3 Caching

System should cache:

- Filing metadata
- Parsed financial statements
- Derived computations

---

14. SECURITY AND PRIVACY CONSIDERATIONS

14.1 Local-Only Data Model

All financial data stored locally.

No mandatory:

- Telemetry
- Cloud sync
- User accounts

14.2 Privacy Principles

Application should:

- Avoid unnecessary outbound requests
- Minimize third-party dependencies

14.3 Secure Storage

Recommended:

- Sandboxed storage
- Integrity-safe local database handling

---

15. FUTURE ENHANCEMENTS

15.1 V2 Enhancements

Custom Metrics & Export *(specified — see `docs/custom-metrics-prd.md`)*

- **Custom metrics (user-imported KPIs)** — per-company, user-supplied business indicators (e.g. PayPal's Total Payment Volume, active accounts), imported via CSV, period-aligned, and charted alongside built-in metrics. User data is held in a separate store and excluded from source lineage.
- **CSV export** — download any chart series or financial table as full-precision CSV (built-in metrics and custom metrics), for validation and sharing. Backend-generated to preserve exactness.

Analytical Enhancements

- Multi-company comparison
- Peer benchmarking
- Multiple predefined formulas

Data Enhancements

- Automatic refresh
- Saved watchlists
- Smarter caching

15.2 V3 Enhancements

Intelligence Features

- News ingestion
- Earnings transcript integration
- AI-assisted exploration

Export Features

- Spreadsheet interoperability
- Chart/image export

15.3 Long-Term Possibilities

Potential future areas:

- Portfolio overlays
- Local LLM integration
- Advanced screening
- Plugin SDK
- Formula editor
- Custom dashboards